



Strong-adhesion and nonfouling self-generating zwitterionic Janus hydrogel paint

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ABSTRACT

Zwitterionic hydrogel paints are promising and ecofriendly materials to prevent biofouling and oil contamination. However, the weak adhesion of hydrogel paints to substrates and tedious fabrication and coating processes limit their practical applications. Herein, a novel self-generating zwitterionic Janus hydrogel paint consisting of a polymer copolymerized by methyl methacrylate (MMA), tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSi), and 3-(methacryloxypropyl) trimethoxysilane (KH570) is reported. The polymer, when coated on a substrate, self-crosslinks into the polymer network and interlinks to the substrate during curing. This polymer decouples polymerization from the crosslinking and interlinking processes. When used underwater, the top layer of the coating rapidly hydrolyzes and self-generates a zwitterionic hydrogel layer, whereas the hydrophobic bottom layer remains unhydrolyzed. Adhesion tests show that this coating has considerably stronger adhesion strength than traditional zwitterionic hydrogel coatings due to the synergistic effect of hydrophobic interactions and interface bonding. Moreover, it has excellent resistance to protein, bacteria (*Escherichia coli*, *Staphylococcus aureus*, and *Pseudomonas* sp.), and diatom (*Navicula incerta*). The self-generating zwitterionic Janus hydrogel coating can also prevent oil contamination. This study serves as a promising avenue for the design and fabrication as well as practical application of self-generating zwitterionic Janus hydrogel paint with strong substrate adhesion.

1. Introduction

Hydrogels are three-dimensional hydrophilic polymer networks that can absorb large amounts of water [1,2]. They are ubiquitous and found in muscles and cartilages as well as xylem and phloem [3,4]. They have excellent properties such as high surface superhydrophilicity, lubricity, biocompatibility, and low Young's modulus [5,6]. Several natural and synthetic hydrogels with different polymer topologies and chemical compositions have been prepared and used in medical engineering, stretchable devices, soft robotics, and marine antifouling [7–9]. However, the high water content in hydrogels results in insufficient bulk mechanical properties and their weak bonding with material substrates [10–12]. Composite hydrogels [10], multinet network hydrogels [13], and supramolecular hydrogels [14] have improved mechanical robustness but low interface bonding strength. To ensure strong adhesion, hydrogels must have strong interaction with material substrates besides high cohesive strength. Achieving these properties is a critical challenge and therefore hinders the practical use of hydrogels as coatings.

Owing to large amounts of water at their interface, the adhesion of hydrogels to target substrates is intrinsically weak [15,16]. Therefore, methods such as surface bridge and surface initiation have been used to coat hydrogel paints on surfaces to strengthen the bonding between the polymer network of hydrogels and target substrates [3,17,18]. The surface bridge method uses bridge molecules, such as (3-aminopropyl) triethoxysilane (APTES), to form covalent bonds with the substrate and hydrogel separately, facilitating their firm bonding [3,19]. However, this method relies on the functional groups in the bridge molecules and controlling the thickness of the coating may be difficult [20]. In the surface initiation method, the hydrophobic initiator is adsorbed onto the target substrate surface via diffusion or using an additional primer [20–22]. The hydrogel precursor is then cured on the treated substrate, allowing it to be firmly bonded to the substrate. Although this method can be used for most substrates, the obstruction of oxygen, dependency on the shaping mold, and the presence of toxic monomers and initiators in surface-initiated free-radical polymerization make its application challenging [23,24]. Wang et al. addressed this issue by developing a

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renatured hydrogel paint using a special adhesive that improved the adhesion of the sprayed dehydrated xerogel particles. Then, xerogel particles were rehydrated into a hydrogel coating [23]. Therein, hydrogel fabrication and substrate surface coating were performed independently, making the process scalable. However, it is also tedious because the use of additional adhesive and the renatured process are indispensable. Suo et al. proposed a hydrogel paint that decoupled polymerization from crosslinking and interlinking of the hydrogel on the substrate surface [24]. Similar to ordinary paints, this hydrogel paint was applied to the target substrate using conventional methods, such as brushing and dipping. During the curing of the paint, the hydrogel coating is crosslinked and a strong bond is formed between the coating and substrate. Thus, this approach is scalable and promising to be used in real-world applications. Nevertheless, the hydrogel coating retains its inherent high hydrophilicity, and the abundant water at the interface significantly weakens the interlinking moieties and prevents the strong bonding of hydrogel with the solid substrate surface. Moreover, the functionality of hydrogel coating must be further improved to facilitate their wide application.

Zwitterionic hydrogels have a dense and stable hydration layer; therefore, they exhibit excellent anti-biofouling properties [25–27] and underwater oleophobicity [28–30]. However, these hydrogels swell in water due to their high hydrophilicity, which impacts their mechanical properties and leads to extremely weak bonding with the target substrates [31,32]. In particular, introducing hydrophobic or non-zwitterionic compounds in these hydrogels is difficult owing to the highly polar nature of zwitterions because they are incompatible and do not have the same solvent. The fabrication and coating of zwitterionic hydrogels on substrates are also complex. Therefore, obtaining zwitterionic hydrogel coatings with strong substrate adhesion is highly desired but a challenging task.

To address these issues, a novel self-generating zwitterionic Janus hydrogel paint with strong substrate adhesion and excellent surface antifouling properties is fabricated herein. A hydrophobic zwitterionic hydrogel precursor is directly obtained by the copolymerization of methyl methacrylate (MMA), a hydrophobic hydrolysis-induced zwitterionic monomer (tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate, TCBSi), and a silane coupling agent (3-(methacryloxypropyl) trimethoxysilane, KH570). The existence of MMA is to improve the film-forming properties and regulate the mechanical performance of the polymeric coating. Carboxybetaine-type zwitterionic polymers are known for their excellent antifouling properties owing to their excellent hydration capacity. Benefiting from the hydrophobicity, the precursor can be widely applied to various substrates without any surface treatment. The silane coupling agents can self-crosslink and bridge the hydrogel coating and substrate. The coating surface is expected to hydrolyze and swell in water to self-generate a thin zwitterionic hydrogel layer with antifouling properties. The unhydrolyzed hydrophobic bottom layer will simultaneously bond with the substrate firmly via hydrophobic interactions. Owing to the hydrophobicity of the bottom layer, water molecules are not trapped between this layer and the substrate [33]. Thus, the interfacial bonding of silane coupling agents is not weakened. Further, this study investigates the surface wettability, bilayer structure, anti-biofouling and anti-oil-fouling performances of the coating. We aim to develop a viable pathway to achieve an anti-fouling self-generating zwitterionic Janus hydrogel paint with strong substrate adhesion.

2. Experimental section

2.1. Materials

Triisopropylsilyl acrylate (TIPSA) was synthesized according to the procedure described in a previous report [34]. Methylaminoethanol, acryloyl chloride, triethylamine, methyl methacrylate (MMA), 3-(methacryloxypropyl) trimethoxysilane (KH570), 2,2-

azobisisobutyronitrile (AIBN), and *n*-hexane were purchased from Aladdin. AIBN was purified by recrystallizing twice in methanol before using. Artificial seawater (ASW) was prepared according to ASTM D1141-98 (2003). Tetrahydrofuran (THF) and dioxane were distilled by CaH₂ before using. Fibrinogen (*M_w* = 340 kDa) was purchased from Merck Chemicals. Physiological phosphate buffered saline (PBS, pH = 7.4) was prepared according to the published article [35].

2.2. Synthesis of tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSi)

TCBSi was synthesized according to the route of our previous report [36]. Briefly, 15 g of methylaminoethanol (0.2 mol) was added dropwise into 45.6 g of triisopropylsilyl acrylate (0.2 mol) at 0 °C under argon atmosphere. After stirring for 2 h, 300 mL of THF, 22.2 g of triethylamine (0.22 mol), and 19.9 g of acryloyl chloride (0.22 mol) were also added into the system dropwise at 0 °C. Then the reaction was stirred for another 6 h. Finally, the mixture was filtered, concentrated, dissolved by ethyl acetate, and extracted by saturated NaCl solution. The target product was obtained after drying under vacuum overnight.

2.3. Synthesis of copolymer

Typically, 1.91 g of MMA, 3 g of TCBSi, 0.14 g of KH570, and 0.04 g of AIBN were dissolved in 10 mL of dioxane. Then the mixture was degassed by three freeze-evacuate-thaw cycles and backfilled with nitrogen. The reaction was kept at 70 °C for 24 h. The resultant copolymer was precipitated in hexane and dried under vacuum at 70 °C overnight. For convenience, the polymeric coatings were termed as MS_xKy, where *x* and *y* refer to the mole ratios of TCBSi and KH570 respectively. ¹H NMR (600 MHz, CDCl₃, ppm, Fig. S2): 4.11 (-CH₂COOCH₂CH₂- from TCBSi), 3.63 (-COOCH₃ from MMA), 3.58 (-SiOCH₃ from KH570), 2.78 (-N(CH₃)CH₂CH₂COO- from TCBSi), 2.71 (-CH₂N(CH₃)CH₂CH₂COO- from TCBSi), 2.30 (-N(CH₃)CH₂CH₂COO- from TCBSi), 1.30 (-SiCH(CH₃)₂ from TCBSi), 1.07 (-SiCH(CH₃)₂ from TCBSi), 0.65 (-CH₂Si(OCH₃)₃ from KH570).

2.4. Characterization

Nuclear magnetic resonance (NMR) spectra were conducted on a Bruker AV600 NMR spectrometer where CDCl₃ was used as the solvent and tetramethylsilane was used as the internal standard.

2.5. Preparation of polymer films

Each (~300 μm) polymer film was prepared by solution casting method. The solution of MS_xKy and THF (30 % w/v) was coated on various substrates and dried at room temperature for 3 days to evaporate the solvent, and the weights of all polymer films reached equilibrium. The thickness of the coating was measured by using a coating thickness gauge (Phynix, Model Surfex). To characterize properties of hydrolyzed coating, the hydrolysis of coating was accelerated in 0.1 M NaOH solution. Considering that the actual application environment of self-generating zwitterionic Janus hydrogel paint is marine, the pH of hydrolysis process was controlled at 8.2 (for long-term stability) or 0.1 M NaOH (for acceleration) and the temperature was kept at 25 °C.

2.6. Rheological test

Dynamic rheology experiments were performed on a rheometer (ARES-G2, TA Instruments) with a recessed concentric cylinder geometry (20 mm diameter) at 25 °C. Flow sweep tests were performed at shear rates ranging from 10⁻¹ to 10³ s⁻¹. Time sweep oscillatory tests were performed at 100 % strain and 6.28 rad/s with a 0.1 mm gap for 600 s.

2.7. Determination of equilibrium water content

The weight of each hydrolyzed and fully swollen coating ($\sim 300 \mu\text{m}$) was recorded as W_S . The weight of the coating after freeze-drying and absorbed water removal was recorded as W_D . The equilibrium water content (EWC) of the coating can be calculated by using the following equation.

$$\text{EWC} = \frac{W_S - W_D}{W_S} \times 100\%$$

2.8. Attenuated total reflection infrared (ATR-IR) spectroscopy

Bruker Vertex 70 Fourier transform infrared spectrometer was used to collect the ATR-IR spectra of the coatings with or without hydrolysis at a spectral resolution of 4 cm^{-1} with 32 scans.

2.9. X-ray photoelectron spectroscopy (XPS)

XPS analysis was performed using an Axis Ultra DLD (Kratos Analytical) with a monochromate Al K α ($h\nu = 1486.6 \text{ eV}$) source at takeoff angles of 90° . C (1 s) at 284.6 eV was used as the reference for the binding energy.

2.10. Water contact angle (WCA) and underwater oil contact angle (OCA) measurement

The values of static WCA and OCA were measured on a theta optical tensiometer (Attension, Biolin Scientific) by dropping a droplet of liquid ($4 \mu\text{L}$) on the surfaces of coatings at 25°C . Flip the coatings and place them underwater to measure OCA. The average value was obtained by measuring five different regions.

2.11. Scanning electron microscopy (SEM)

The structure of the coating surface and cross-section was observed using a scanning electron microscope (ZEISS MERLIN) equipped with an energy dispersive X-ray spectrometer (EDS).

2.12. Adhesion test

The adhesion strength of the coatings on diverse substrates was measured by using a pull-off adhesion tester (PosiTest AT-A Automatic). The adhesion strength data were obtained by detaching an aluminum dolly (2.0 cm in diameter) at a speed of 0.2 MPa/s at 25°C . Five parallel tests were conducted to obtain an average value.

2.13. Quartz crystal microbalance with dissipation (QCM-D)

The protein resistance ability of MSxK2 coatings was investigated by QCM-D from Q-sense AB (Sweden). The coatings were prepared by spin-casting with precursor solution in THF (5.0 mg/mL) on an AT-cut quartz crystal and cured at 25°C for 24 h. Fibrinogen solution (1.0 mg/mL of ASW) was delivered to the coating surface at a flow rate of $150 \mu\text{L/min}$ with ASW as reference. Briefly, the frequency change (Δf) is related to the protein adsorption on the coating surface, while the dissipation change (ΔD) is correlated to the viscoelastic properties of the adsorbed protein layer. More details of the QCM-D measurement can be found elsewhere [37,38].

2.14. Antibacterial assays

Gram-positive bacteria (*S. aureus*), gram-negative bacteria (*E. coli*), and marine bacteria (*Pseudomonas* sp. NOV776) were used to evaluate the antibacterial performance of the MSxK2 coatings. The coatings were prepared on silicon wafers ($1 \times 1 \text{ cm}^2$) by spin-casting with a 24 h curing

process at 25°C , and immersed in 0.1 M NaOH solution for 5 min to pre-hydrolyze. A silicon wafer without any coating was used as the blank control. Then, all the silicon wafers were immersed in the bacterial suspension (10^7 CFU/mL) of ASW (for *Pseudomonas* sp.) or PBS (for *S. aureus* and *E. coli*) for 4 h at 30°C , and slightly rinsed with ASW or PBS to remove the unattached bacteria. Subsequently, the silicon wafers were treated by ultrasonication for 10 min to collect the strongly adhered bacteria. $100 \mu\text{L}$ of the collected suspension was spread onto the agar plates and further incubated at 30°C for 24 h. Specially, the antibacterial performance of the bottom layer of MS30K2 was also tested by peeling it from the tinplate after hydrolysis.

2.15. Anti-diatom test

The diatom *Navicula incerta* (*N. incerta*) obtained from Xiamen University was cultivated in an F/2 culture medium under a 12:12 h light-dark cycle at 23°C in a light incubator. The copolymer coating was also prepared on a silicon wafer ($1 \times 1 \text{ cm}^2$) by spin-casting with a 24-h curing process at 25°C and immersed in 0.1 M NaOH solution for 5 min to pre-hydrolyze. A blank silicon wafer was chosen as the control sample. The sample was placed in a 24-well microplate and immersed in diatom suspension (a chlorophyll value of $0.25 \mu\text{g/L}$). After soaking for 24 h, ASW was used to rinse the sample, and the adhered diatoms were observed by using a microscope (Scope A1, Zeiss). Similarly, MS30K2 coating was peeled from the substrate after hydrolysis and characterized the anti-diatom property of the bottom layer.

2.16. Anti-oil exhibition

The model boat was made of art paper and covered by aluminum tape. The solution of the uncrosslinked coating precursors was coated onto the model boat directly. After the coating was cured, the model boat was immersed in water completely to hydrolyze before use and then placed in water covered by a layer of mineral oil (dyed orange).

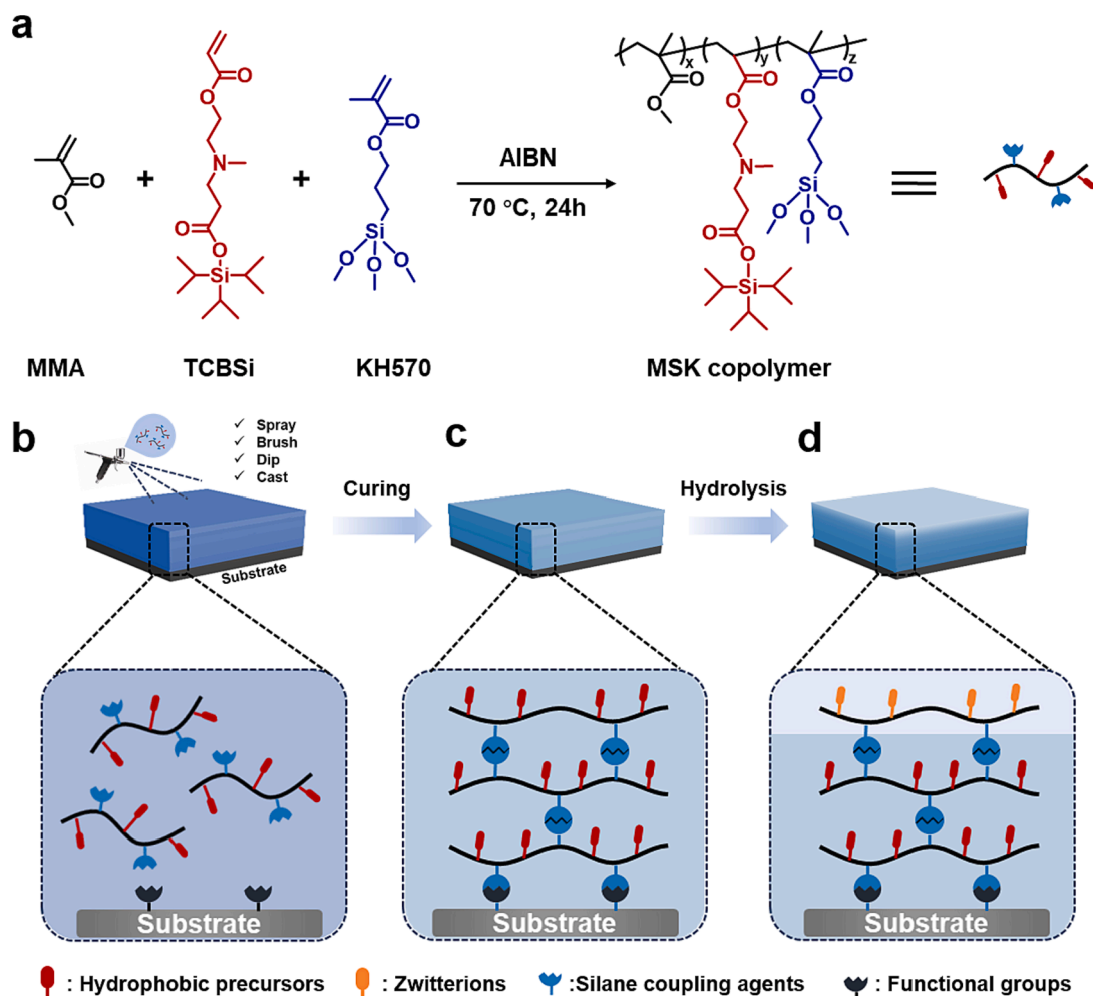
2.17. Statistical analysis

All outcomes are expressed as means $\pm$ standard deviation (S.D.). A SPSS 26.0 statistical software was used for statistical analysis. For antibacterial and anti-diatom tests, one-way ANOVA was used to determine differences among groups. P -value < 0.05 was considered statistically significant.

3. Results and discussion

The high hydrophilicity and strong polarity limit the application of zwitterionic hydrogels as coatings. Therefore, hydrophobic hydrolysis-induced zwitterionic monomers were fabricated herein, which hydrolyzes only when contacted with water and can regulate the degree of hydrolysis [31,32,36]. In addition, the hydrophobic zwitterionic monomers hydrolyzes in ASW and releases the siloxane product which is not toxic for the marine organisms [39–41]. These hydrophobic precursors can copolymerize with other hydrophobic or non-zwitterionic compounds and easily dissolve in common solvents. They can also be coated on various substrates without water molecules at interface, wherein they undergo hydrophobic interactions to ensure adhesion [42–44]. To separate the hydrogel fabrication and substrate surface coating processes, a silane coupling agent was introduced. This agent underwent self-condensation during curing, ensuring strong interaction between the coating and the substrate interlinked by silane coupling agents and improved adhesion without the disturbance from water at interface [24].

Scheme 1a shows the preparation of the hydrogel precursors of MMA, TCBSi, and KH570. Their successful syntheses were confirmed by the disappearance of the peaks assigned to the protons of $\text{C}=\text{C}$ at $6.3\text{--}5.8 \text{ ppm}$ in ^1H NMR spectrum. Details can be found Fig. S1 and S2.



Scheme 1. (a) Synthesis routes of MSxKy. (b) Precursors of MSxKy applied to the substrate using various approaches. (c) Silane coupling agents crosslinked for curing and bonded to the substrate surface for improved adhesion. (d) Schematic of self-generation of the hydrogel layer: the top layer hydrolyzes and the bottom layer remains unhydrolyzed.

For convenience, the hydrogel precursors were termed as MSxKy, where x and y refer to the molar ratios of TCBSi and KH570, respectively. Owing to its good solubility in organic solvents and post-curing properties, the precursor solution containing uncrosslinked polymers can be coated to the substrate using various approaches (Scheme 1b). After painting, the silane coupling agents condensed and crosslinked with each other to form a polymer network, which also bonded to the functional groups on the substrate (Scheme 1c). After curing, the coating surface hydrolyzed in water to self-generate zwitterions and then swelled up and absorbed water to form a zwitterionic hydrogel layer (Scheme 1d). The self-generating Janus hydrogel coating obtained using hydrophobic hydrolysis-induced zwitterionic monomer and the silane coupling agent can enhance the bulk mechanical properties and improve the interfacial adhesion of traditional hydrogels.

To regulate the hydrolysis degree and swelling of the coating and the self-generation of the hydrogel layer, the molar ratios of TCBSi and KH570 in the MSxKy system must be optimal. To determine the optimal ratio, different coatings with varying TCBSi and KH570 contents were prepared. The KH570 content was 2 mol% based on the data (Fig. S3 and S4). In terms of the TCBSi content, the initial water contact angles (WCAs) slightly increased with TCBSi content containing triisopropylsilyl ester as the terminal group (Fig. 1a). After hydrolysis, the WCAs of all samples decreased, except for MS0K2, with increasing TCBSi content. Similarly, all the coatings showed initial oil contact angles (OCAs) of $\sim 100^\circ$, which increased after hydrolysis, except MS0K2,

indicating that the surface hydrolyzed and generated more hydrophilic and oleophobic zwitterions (Fig. 1b). MS0K2, MS10K2, and MS20K2 maintained their dense structures (Fig. 1c). Only MS30K2 swelled in the aqueous environment and showed a porous structure. Their equilibrium water contents also sharply increase with the TCBSi content (Fig. S5). When the TCBSi content is 30 mol%, the coating with an equilibrium water content of 49.2 % became sufficiently hydrophilic to form a hydrogel layer. At a higher TCBSi content (~ 40 mol%), the coating swelled quickly and severely such that it easily detached from the substrate. Thus, coatings containing 2 mol% of KH570 and 30 mol% of TCBSi swell appropriately to self-generate a hydrogel layer.

When the top layer of the coating hydrolyzed and self-generated a hydrogel layer, its bottom layer remained unhydrolyzed, which could help maintain strong adhesion. Thus, the coating may have a bilayer structure, which was confirmed by characterizing the structure of MS30K2. As shown in Fig. 1d–1f, the top layer of MS30K2 coating is porous and oleophobic, whereas the bottom layer is dense and oleophilic, indicating that the bottom layer remains unhydrolyzed when the top layer hydrolyzes and forms the zwitterionic hydrogel layer. Fig. 1e shows the energy-dispersive X-ray spectroscopy (EDS, line scan) analysis of Si distributed at the cross-section of the zwitterionic hydrogel coating. Si from TCBSi found in the top layer is lower than that in the bulk and bottom layers of the coating, indicating that only the TCBSi in the top layer hydrolyzes and the terminal group of triisopropylsilyl ester leaves from the coating. The bilayer structure and hydrolysis were further

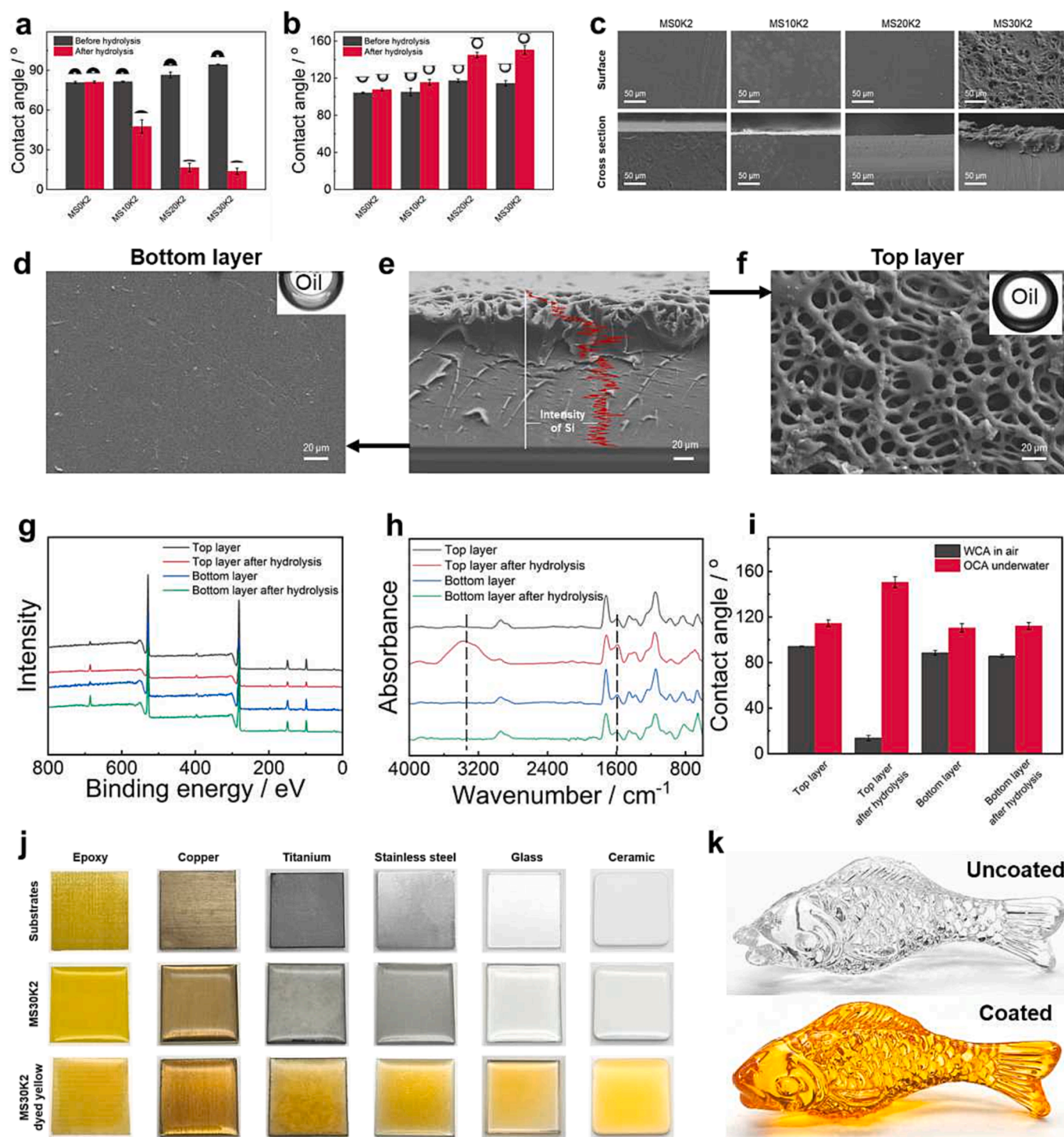


Fig. 1. (a) Water contact angles (WCAs) in air and (b) oil contact angles (OCAs) underwater of MSxK2 coatings with different TCBSi contents before and after hydrolysis. (c) SEM images of the cross-section and surface of the MSxK2 coating after hydrolysis. SEM images and OCAs of the (d) top layer and (f) bottom layer of MS30K2 hydrogel coating. (e) EDS (line scan) spectra of the cross-section of the MS30K2 hydrogel coating. (g) XPS spectra, (h) ATR-IR spectra, and (i) contact angle of the top layer before and after hydrolysis and bottom layer before and after hydrolysis. (j) MS30K2 coated on various substrates. (k) Appearance of the glass fish coated with MS30K2.

analyzed via X-ray photoelectron spectroscopy (XPS), attenuated total reflection infrared (ATR-IR), and contact angle measurement. As shown in Fig. 1g, the peaks at 150.5 and 100.4 eV are attributed to Si in TCBSi (2s and 2p). After hydrolysis, the Si peak intensity of the bottom layer hardly changes but that of the top layer weakens. ATR-IR spectra show similar results (Fig. 1h). After hydrolysis, new peaks for hydroxyl and carboxyl groups can be observed at 3387 and 1576 cm^{-1} , respectively,

and the peak for Si-C at 885 cm^{-1} disappears. Moreover, the WCAs and OCAs of the top and bottom layers decrease and increase, respectively, after hydrolysis (Fig. 1i). These results confirm the bilayer structure of the MS30K2 hydrogel coating. Note that the hydrogel layer is about 40 μm according to Fig. 1e. That is, the theoretical minimum thickness of the coating to achieve Janus structure should be higher than 40 μm .

To verify that the polymeric precursor solution is uncrosslinked and

can be coated to the substrate using various approaches, the rheological behavior of the hydrogel precursor solutions was tested. As shown in Fig. S6a, the loss modulus (G'') is always higher than the storage modulus (G') over the storage time, and the values almost do not change. Moreover, the precursor solution also keeps at a low-viscosity state over the storage time (Fig. S6b). These results indicate that the polymeric precursor solution is uncrosslinked and has good stability. As shown in Fig. 1j and 1k, MS30K2 was easily and successfully coated on a wide range of substrates, including irregularly shaped substrates, without any additional surface treatment. This indicates that self-generating zwitterionic Janus hydrogels can be coated on complex and fine geometrical substrates.

In general, the weak adhesion of hydrogel coatings caused by abundant water molecules in the network and at the interface has seriously hindered their widespread application. MS30K2 Janus hydrogel coatings are expected to combine high antifouling capacity and strong adhesion in the top and bottom layers. As shown in Fig. 2a, due to the lack of hydrophobic interaction and covalent bonding, traditional zwitterionic hydrogel coatings have extremely weak adhesion (~ 0.13 MPa). MS30K2 hydrogel coating shows considerably strong adhesion to diverse substrates: 2.56 MPa for epoxy, 3.17 MPa for copper (Cu), 3.34 MPa for titanium, 3.70 MPa for stainless-steel, 5.54 MPa for glass, and 7.52 MPa for ceramic. The high polar interaction and strong hydrophobic interaction between the coating and epoxy or metal substrates result in superior adhesion; thus, these coatings can be used in applications operating in harsh environments such as marine environments. In addition, the strong and stable interfacial covalent bonds derived from the interlinking siloxane between the coating and glass or ceramic further improve the adhesion in the absence of water. After hydrolysis in 0.1 M NaOH solution, the adhesion strengths of the MS30K2 coating on different substrates slightly decrease because the hydrolysis occurs only on the top layer of the coating. As shown in Fig. 2b, the joints of the above substrates can easily withstand a 20 kg kettlebell. The long-term

stability and corresponding changes in the coating's microstructure were evaluated. As the immersion time in ASW increases, coating surface gradually becomes rough and porous due to the swelling while the bottom layer remained dense, also indicating that the hydrolysis occurs only on the top layer (Fig. S7). Moreover, the hydrogel precursors of MMA, TCBSi, and KH570 do not have degradable units and they can self-crosslink into the stable three-dimensional polymer network and interlink to the substrate during curing. Thus, the adhesion of the coating only changes slightly (Fig. 2c and Fig. S8b). On the other hand, the influence of thicknesses on the water contact angle and adhesion strength of MS30K2 coatings was also evaluated. Fig. S8 shows that the coatings with different thicknesses have the same hydrophilicity and mechanical performance before and after hydrolysis in 0.1 M NaOH. The results discussed above demonstrate that the self-generating zwitterionic Janus hydrogel paint possesses relatively good long-term stability. In addition to the pull-off adhesion test, the coating's resistance to water impact was tested (Fig. 2d and Movie S1). MS30K2 and traditional zwitterionic hydrogel coating were impinged by a water jet with a pressure of ~ 2 MPa. Results revealed that the MS30K2 surface was not damaged, whereas the traditional zwitterionic hydrogel coating was completely washed off by the water jet. Thus, MS30K2 has high and stable adhesion strength and can be applied to various substrates.

Protein adsorption is the first step of biofouling; therefore, protein resistance is crucial for marine antifouling [45]. Protein adsorption onto the MSxK2 coating was evaluated by quartz crystal microbalance with dissipation (QCM-D) measurements. Fibrinogen was used as the model protein owing to its strong adsorption ability. MSxK2 Janus hydrogel coatings contain zwitterions; therefore, they are expected to exhibit good anti-biofouling performance. Fig. 3a shows the time dependence of the frequency shift (Δf) and energy dissipation shift (ΔD) for the adsorption of fibrinogen on various MSxK2 surfaces. For MS0K2, Δf decreases and ΔD increases sharply after the introduction of fibrinogen, indicating high protein adsorption. After the MSxK2 surface was rinsed

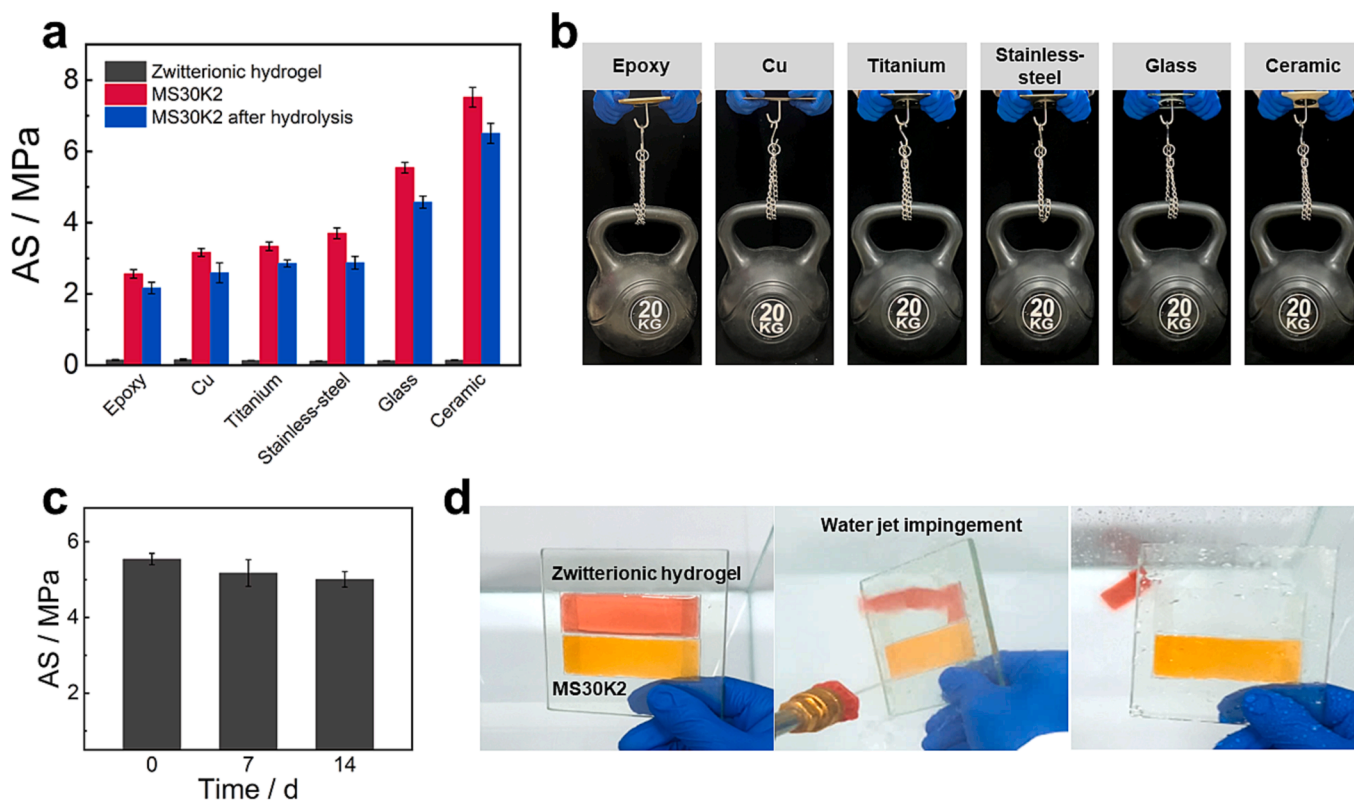


Fig. 2. (a) Adhesion strength of the Janus hydrogel coating on different substrates. (b) Photographs of MS30K2 bonded to the substrates subjected to the load from a 20-kg kettlebell. (c) Adhesion strength of MS30K2 (for glass) immersed in ASW over time. (d) Water jet impingement test of MS30K2 (water pressure ~ 2 MPa).

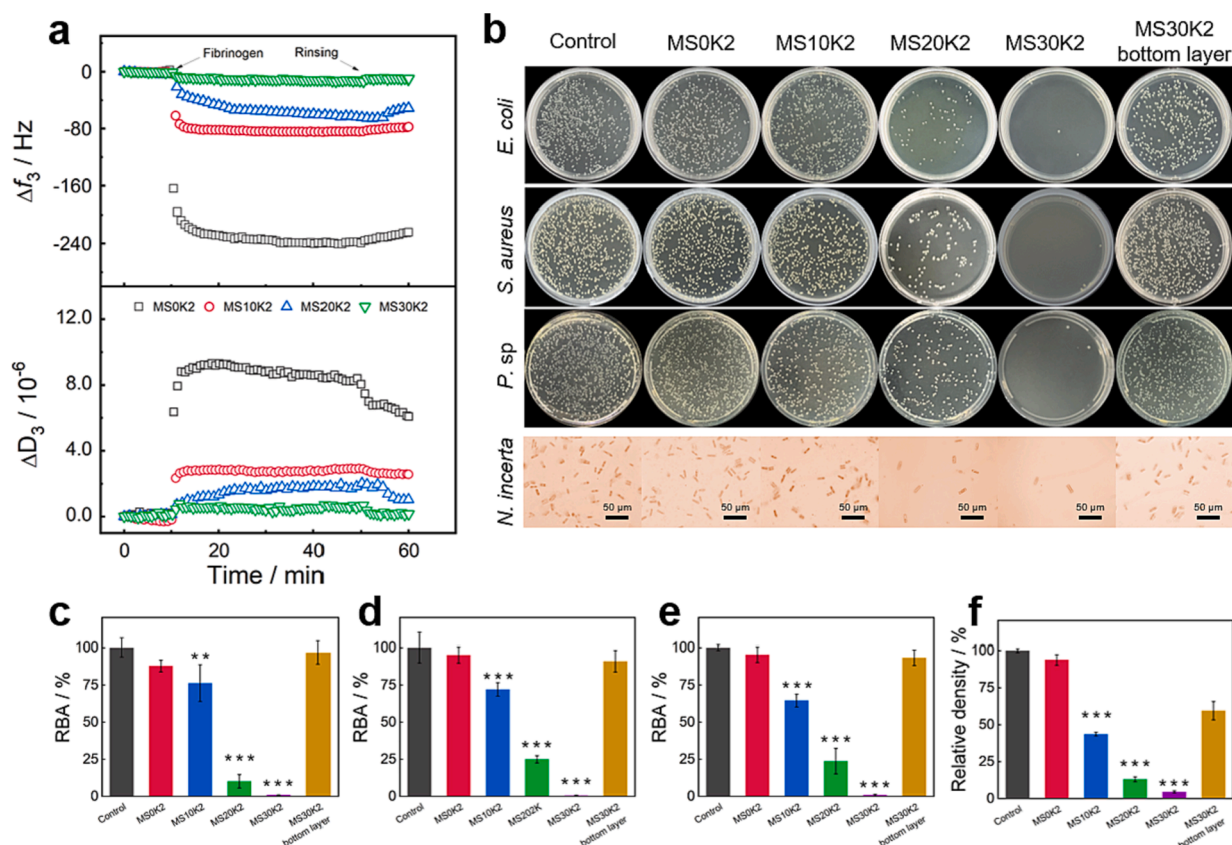


Fig. 3. (a) Time dependence of the frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of fibrinogen on MSxK2 coatings. (b) Agar plate photographs of bacterial colonies (*E. coli*, *S. aureus*, and *Pseudomonas* sp.) and microscopic images of the diatom (*N. incerta*) after incubation with MSxK2 coatings. (c–e) RBA of *E. coli*, *S. aureus*, and *Pseudomonas* sp., respectively. (f) Relative density of *N. incerta*.

with ASW, a notable increase in Δf and a decrease in ΔD are observed, indicating that a viscoelastic layer was formed due to protein adsorption. As the TCBSi content increases, the change in Δf and ΔD decrease, indicating an improvement in the protein-resistance ability of the coating. The changes in Δf and ΔD for MS30K2 after the introduction of fibrinogen are almost negligible because of the formation of a zwitterionic hydrogel layer on the coating, which effectively resists protein adsorption.

Fig. 3b shows the typical optical photographs of agar plates dispersed by suspension of gram-positive bacteria (*Staphylococcus aureus*), gram-negative bacteria (*Escherichia coli*), and marine bacteria (*Pseudomonas* sp.) and the microscopic images of the diatom (*Navicula incerta*) after incubation with MSxK2 coating. Fig. 3c–3f show their relative bacterial adsorption (RBA) and relative density of diatom. A large number of bacterial colonies were observed on the plate of blank control and MS0K2, indicating no resistance to bacterial adhesion. Contrarily, considerably fewer bacterial colonies were observed on the plate of MS10K2 ($p < 0.01$ for *E. coli* and $p < 0.001$ for other bacteria), indicating that MSxK2 coating possessed antibacterial capability even at low zwitterionic content. The number of bacterial colonies decreased as the TCBSi content increased. There were almost no bacterial colonies on the MS30K2 plate ($p < 0.001$), indicating its superior antibacterial property. Particularly, a large number of bacterial colonies were observed on the plate of the bottom layer of the MS30K2 coating, indicating that the bottom layer had not yet hydrolyzed; this also confirms the bilayer structure of the MS30K2 coating. MS30K2 coating exhibited excellent antibacterial activity against the three types of bacteria. In other words, the hydration layer formed by zwitterions possesses a broad spectrum of antibacterial activities.

As a representative fouling organism that accumulates easily on submerged surfaces, a diatom was also used to evaluate the anti-

biofouling properties of the MSxK2 coating. A blank silicon wafer was chosen as the control. As diatoms tend to settle on hydrophobic surfaces, a large number of diatoms were observed on the silicon wafer surface. The surface of MS0K2 coating with similar hydrophobicity to the silicon wafer was also covered by a large number of diatoms ($\sim 94\%$, $p < 0.05$). As the TCBSi content increased from 0 to 30 mol%, the relative density of diatoms on the coating surface decreased from $\sim 94\%$ to $\sim 5\%$. This is because the top layer of coating hydrolyzed to generate a large number of zwitterions, which could effectively inhibit the attachment of diatoms. Notably, the diatom density at the bottom layer of MS30K2 coating was higher than that at the top layer but lower than MS0K2, indicating that the bottom layer of MS30K2 coating was partially hydrolyzed. This layer was slightly hydrolyzed and generated a few zwitterions after incubation with diatom suspension for 24 h in an aqueous environment.

The anti-oil-fouling surface plays a crucial role in antifouling, environmental management, and oil recovery [46–48]. To verify the anti-oil-fouling properties of MS30K2 coating, a wide range of oils was used. We first evaluated the dynamic behavior of dichloroethane droplets on MS30K2 coating under water. As shown in Fig. 4a, dichloroethane droplets (dyed orange) rolled off smoothly from the tilted glass slide coated with MS30K2 after hydrolysis, leaving no residue compared to the unhydrolyzed coating. This is because MS30K2 coating surface hydrolyzes and self-generates a zwitterionic hydrogel layer, which is hydrophilic enough to form a dense and stable hydration layer that effectively prevents the adhesion of oils. Besides, another five types of common oils including coal tar, motor, pump, cooking, and mineral oil were also tested. Fig. 4b shows that the contaminated stainless-steel plates half-coated by MS30K2 were immersed in water and shaken slightly. Then, the oils easily fell off the MS30K2 coating, while that on the uncoated region were barely removed. That is, MS30K2 coating

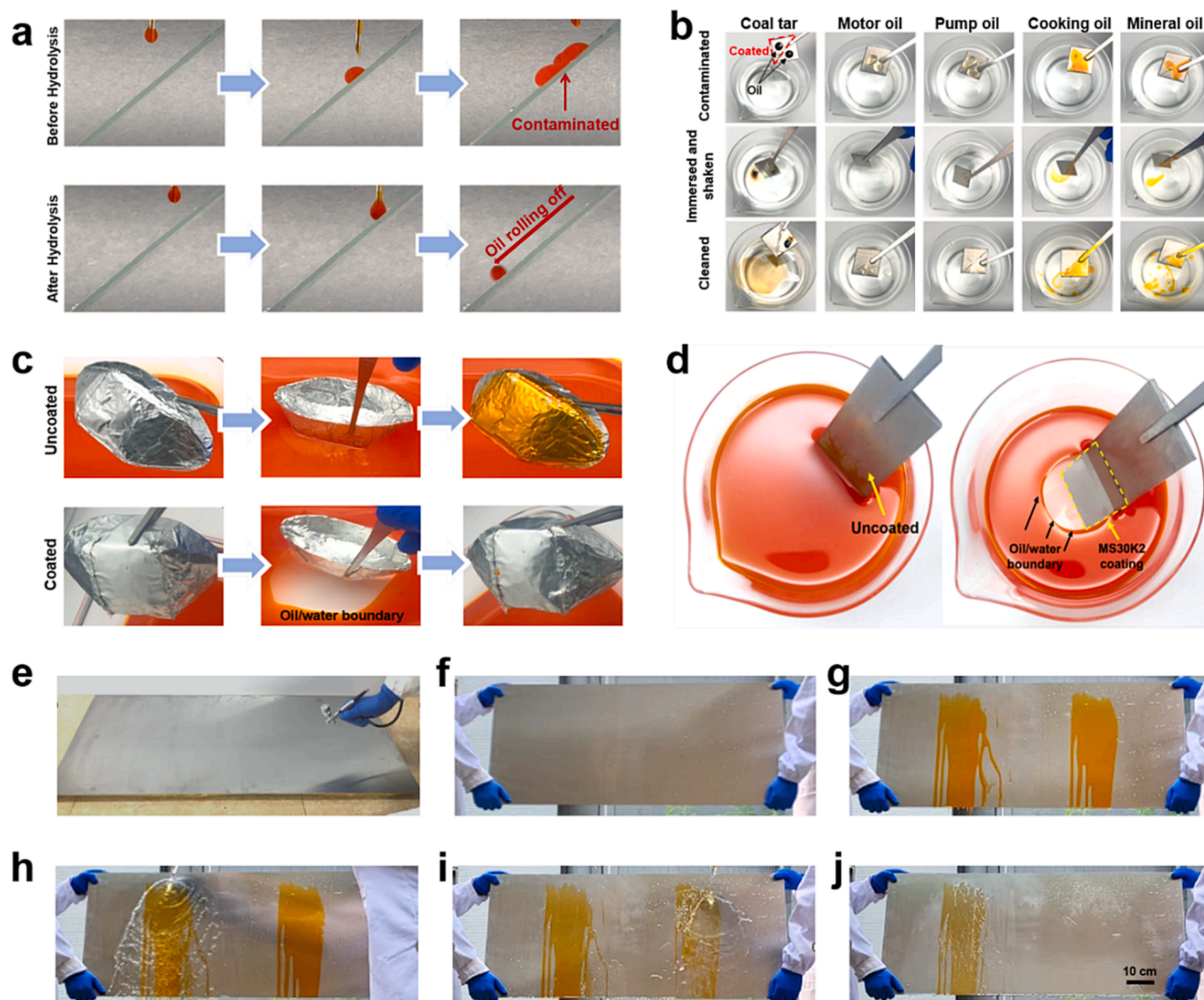


Fig. 4. Photographs of (a) Resistance of MS30K2 to dichloroethane before and after hydrolysis. (b) stainless-steel plate contaminated with oil, immersed in water, shaken and cleaned. (c) Aluminum model boat coated with MS30K2 placed in water covered by a layer of mineral oil. (d) Oil-repellent experiment of MS30K2. (e) Demonstration of the MS30K2 spraying process. (f) Photograph of untreated (left) and MS30K2-painted (right) stainless-steel plate ($0.5 \times 1.0 \text{ m}^2$). (g) Surface contaminated with oil (dyed orange). (h, i) Surface rinsed with a stream of water. (j) Anti-oil-fouling effect after rinsing. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

possesses a broad spectrum of anti-oil-fouling abilities. For more visual presentation (Fig. 4c and Movie S2), an aluminum model boat was made, coated with MS30K2, and placed in water covered by a layer of oil (dyed orange). It could repel the oil, form an oil–water interface, and show a clean bottom after taken out from water. On the contrary, the uncoated boat was closely contacted by the oil and severely contaminated. Similarly, the oil-repellent phenomenon can also be observed obviously when the stainless-steel plate coated with MS30K2 coating inserted in water with oil layer (Fig. 4d). All the above results indicate that MS30K2 coating has excellent anti-oil-fouling properties owing to the formation of a zwitterionic hydrogel layer. In particular, the fabrication of self-generating zwitterionic Janus hydrogel paint is time- and labor-saving, it can be easily applied to various substrates on a large scale. Herein, it was sprayed on a stainless-steel plate ($0.5 \times 1.0 \text{ m}^2$) to verify its potential for large-scale use (Fig. 4e and 4f). After contaminated with mineral oil and rinsed with water, the anti-oil-fouling ability of MS30K2 was convinced by the obvious cleaner surface on the painted side (Fig. 4g–4j and Movie S3).

4. Conclusion

In this study, a self-generating zwitterionic Janus hydrogel coating was prepared by synthesizing a copolymeric precursor containing a hydrophobic hydrolysis-induced zwitterionic monomer and silane coupling agent. This coating could be applied to various substrates on a large scale and shaped in one step using different coating methods such as spraying, brushing, dipping, and casting. On hydrolysis, the coating formed a bilayer structure. The top layer of the coating self-generated a zwitterionic hydrogel layer that exhibited excellent resistance to protein and oil as well as anti-biofouling performances against gram-positive bacteria (*S. aureus*), gram-negative bacteria (*E. coli*), marine bacteria (*Pseudomonas* sp.), and diatom (*N. incerta*). The bottom layer of the coating was unhydrolyzed and eliminated the effect of water at the interface, ensuring a significantly higher adhesion strength than traditional zwitterionic hydrogel coatings. Using the proposed novel approach, zwitterionic hydrogels with strong substrate adhesion and excellent surface antifouling properties can be fabricated, which can be applied on an industrial scale to medical and marine fields.

CRediT authorship contribution statement

Yichen Zhang: Writing – original draft, Validation, Investigation.
Qingyi Xie: Supervision. **Jiansen Pan:** Writing – review & editing, Supervision. **Chunfeng Ma:** Writing – review & editing, Supervision, Conceptualization. **Guangzhao Zhang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2024.149959>.

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